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Does the depth of digital trade rules promote bilateral value chain cooperation?

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ABSTRACT

This study explores the impact of the depth of digital trade regulations on bilateral value chain (BVC) cooperation within the framework of regional trade agreements (RTAs) using the University of International Business and Economics global value chain database and the Trade Agreements Provisions on Electronic Commerce and Data database from 2001 to 2014. The number of digital trade provisions in RTAs has been increasing since the US–Jordan RTAs first included them in 2000, and countries are paying more attention to digital trade negotiations. The author has constructed indicators of bilateral digital rules and classified them. At the sector level, indicators of the degree of BVC cooperation were also constructed. The result confirms that the depth of digital trade rules primarily facilitates BVC cooperation through four pathways, including lower bilateral trade costs, investment effects, strengthening spatial knowledge spillovers, and optimizing bilateral business environments. Furthermore, the heterogeneous promotion effects of digital trade rules on BVC cooperation were analyzed from four perspectives (i.e., type of country pairs, regional differences, industry differences, and types of provisions). The results of this analysis are significant and hold for various robustness checks.

1. Introduction

Global attention has focused on the gradual emergence of digital technology as a new driver and engine of global economic growth. At the micro level, digital technologies have facilitated the upgrading of consumer consumption (Xu, Guo, Yue, & Li, 2024; Ye & Yue, 2024; Yue, Korkmaz, Yin, & Zhou, 2022; Zhou, Shi, Bao, & Gao, 2023) and increased labor demand (Liang, Liang, & Tang, 2024). In international trade, digital technologies fundamentally alter the international trade landscape (Ahmed, 2019). This type of technology affects trade in three ways, by broadening its scope, increasing its scale, and increasing its speed (González & Jouanjean, 2017). Data flows have recently gained explicit recognition in trade policy circles as a critical component of social production (Burri & Polanco, 2020). Digital technologies have significantly affected global value chains (GVCs) and economic growth. They have reduced the cost of information and entry barriers to international trade, providing new opportunities for many small- and medium-sized enterprises to grow internationally and participate in the GVC (González & Ferencz, 2018; González & Jouanjean, 2017).

Digital technologies have also posed unprecedented challenges, with some contentious "frontier issues" impeding the positive effects of digital trade. On the one hand, digital trade raises several regulatory concerns (Ahmed, 2019). The development of the global digital trade has been limited by significant issues (e.g., cross-border data flows, localization of data storage, protection of personal privacy, and intellectual property rights protection). The free flow of cross-border data, localization of data storage, and protection of individual privacy have been vital points of contention between the United States and the European Union (EU) (Meltzer, 2014). For instance, the US–Mexico–Canada Agreement (USMCA) prohibits data localization storage requirements. Nevertheless, the EU's new General Data Protection Regulation outlines EU privacy standards and the circumstances under which personal data can be processed, whereas personal data cannot be transferred across international borders to jurisdictions with lower privacy protection than the EU (Meltzer, 2019) or exported unilaterally, and EU law is enforced (Elsig & Klotz, 2021). Furthermore, the parties have not agreed on

Abbreviations: BVCs, Bilateral Value Chains; CPTPP, Comprehensive and Progressive Agreement TPP; CRS, Computer and related services; EU, European Union; FDI, Foreign Direct Investment; FE, fixed effects; GVCs, Global Value Chains; KORUS, US–Korea Free Trade Agreement; RTA, Regional Trade Agreements; TAPED, Trade Agreements Provisions on Electronic Commerce and Data; UIBE, University of International Business and Economics; USMCA, US–Mexico–Canada Agreement; WGI, Worldwide Government Indicators; WTO, World Trade Organization.

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permanent exemptions from customs duties for electronic transfers. On the other hand, there is no common definition of digital trade, which the World Trade Organization (WTO) loosely defines as "the production and sale of commodities or services by electronic means." However, in its third definition of "digital trade," the United States International Trade Commission[1] classified digital goods and services into six categories.[2] The scope of the term "digital trade" was expanded, and the importance of cloud computing services in digital trade was stressed. However, the classification of digital trade as part of the trade in goods or services is open to debate.

As countries are already blocking international data flows and mandating data localization, which undermines the economic benefits of digital trade, a framework for digital trade governance is urgently required (Meltzer, 2019). However, the regulation of digital trade through global agreements has made little progress. The WTO Doha Round's implosion has prompted countries to focus on interregional trade rule discussions, including digital trade negotiations (Elsig & Klotz, 2021). Developed economies (e.g., the US and the EU) are actively leading the development of digital trade rules through regional trade agreements (RTAs) to use trade agreements to address the important new issue of digital trade (Gao, 2018). Furthermore, more countries are starting to focus on bilateral negotiations to address a more integrated, digitally driven global economy (Sen, 2018). Currently, the Trans-Pacific Partnership (TPP)[3] represents a higher level of trade and investment standards established by industrialized countries, with more comprehensive digital trade laws. Important new digital provisions in the Comprehensive and Progressive Agreement TPP (CPTPP) and USMCA include prohibitions on data localization, which require parties to respect privacy and encourage best efforts to improve interoperability with domestic privacy regimes.

As digital technologies accelerate the digitization of services and improve their tradability, countries increasingly focus on implementing high-standard digital trade rules to promote competitiveness in the digital economy. The new generation of digital trade rules is at the center of several policy and research debates, as researchers attempt to assess their economic effects and provide guidance on designing and implementing them efficiently. Furthermore, the coronavirus pandemic and increased trade protection threaten GVC production cooperation. As a result, this study seeks to determine whether high-level digital trade regulations can foster and sustain GVC collaboration and intrinsic impact mechanisms.

This study contributes to this broader debate on high-level digital rules by empirically investigating the relationship between high-level digital rules and bilateral value chain (BVC) cooperation from a global standpoint. Using a new data set on the content of digital trade rules provided by the Trade Agreements Provisions on Electronic Commerce and Data (TAPED) database[4] and the University of International Business and Economics (UIBE) GVC database, this analysis allows the quantification of the impact of high-level digital trade rules on BVCs among member countries, as well as sheds light on the mechanism by which high-level digital trade rules affect BVC cooperation. According to the study, signing high-level digital trade rules positively affect BVC cooperation. The positive impact remains significant even after accounting for a wide range of robustness tests. In-depth research reveals that the complexity of digital trade regulations considerably facilitates BVC cooperation in core services. Furthermore, the digital trade rules agreed upon by North–North and North–South countries significantly contribute to BVC cooperation. Although privacy protection and cybersecurity-related provisions impede BVC cooperation, trade promotion and data-related provisions positively affect BVC cooperation. Furthermore, this study suggests that high-level digital trade regulations can promote BVC cooperation via the "trade cost reduction effect," "investment effect," "spatial knowledge spillover effect," and "business environment improvement effect."

In the econometric analysis, we use a sector-level structural gravity model to estimate the relationship between BVC cooperation and the depth of digital trade provisions. To control for the selection bias, we add importer–exporter dimension control variables, importer–exporter–time dimension control variables, importer–time fixed effects (FE), exporter–time FE, and sector–time FE to the baseline model. The TAPED database serves as the foundation for the digital trade rules' depth measures. In the mechanism test, we use the baseline model's interaction terms for digital trade rules depth and mediating variables to analyze the four mechanisms: trade cost reduction effect, investment effect, spatial knowledge spillover effect, and business environment improvement effect.

The remainder of the paper is structured as follows. Section 2 presents the literature review and theoretical mechanism. Section 3 describes the econometric specification and variable construction. Section 4 shows data descriptions. Section 5 focuses on empirical analysis and robustness testing. Section 6 presents mechanism analysis. Finally, Section 7 concludes the paper.

2. Literature review and theoretical mechanism

2.1. Review of the literature

The RTAs have emerged as the best option in a multilateral framework when nations cannot agree on fundamental issues concerning digital trade for an extended period. Digital trade regulations within the context of RTAs can quickly resolve issues that a few nations cannot agree on in the multilateral system, affecting bilateral trade.

The following are the two main types of literature relevant to this article. The first research papers examine the relationship between RTAs and GVCs. Early studies on the relationship between RTAs and import and export trade focused on total trade value. According to Baier and Bergstrand (2007), signing an RTA can significantly increase the contracting parties' bilateral trade volume. Horn, Mavroidis, and Sapir (2010) creatively classified the terms of current RTAs into WTO plus (commitments that go beyond existing WTO obligations) and WTO-X (commitments not covered by the WTO framework), and subsequent researchers built on this work (Kohl, Brakman, & Garretsen, 2016) to

[1] In 2017, the United States International Trade Commission defined digital trade for the third time in its 2017 report to the Office of the US Trade Representative (USTR), Global Digital Trade – Market Opportunities and Major Foreign Trade Barriers.

[2] Internet infrastructure and network communication services are divided into six categories: cloud computing services (e.g., digital processing, storage, analysis, and software applications); digital content, search, and news; e-commerce, payment, and bookkeeping; industrial applications of digital technology; and consumer communication services and connectivity devices.

[3] Before the United States withdrew in January 2017, the TPP, a 12-nation trade agreement, added a substantial new section known as the e-commerce chapter. The CPTPP incorporated the original TPP e-commerce chapter, from which the remaining 11 countries re-signed.

[4] The 2021 version of the TAPED database contains 187 RTAs, the first of which was signed in 2000 and went into effect in 2001. The TAPED database divides digital trade-related provisions into six major categories: electronic commerce, data flows, data transfers in the services chapter, intellectual property chapters, specially formulated agreements, and rules for trade in goods related to big data.

analyze the effect of trade agreement depth on bilateral trade. Academics started to examine the impact of RTA depth on GVCs from a value-added trade perspective with the improvement of the value-added trade accounting system (Hummels, Ishii, & Yi, 2001; Koopman, Wang, & Wei, 2014; Wang, Wei, & Zhu, 2013). Orefice and Rocha (2011) found that RTA integration promotes trade with added value. Meanwhile, according to Laget, Osnago, Rocha, and Ruta (2020), deep RTAs encourage value-added trade in intermediate goods and improve value chain collaboration among nations in two dimensions: forward and backward connections in GVCs (Boffa, Jansen, & Solleder, 2019). These studies analyze the impact of RTAs on both total trade value and value-added trade. Drawing on previous research that examines trade activities through the lens of value-added trade, this article focuses on BVC cooperation between countries. BVCs, as opposed to GVCs, refer to value chain linkages between two countries in which products or intermediate goods from one country are further processed or enhanced in the other before being exported or entered into other markets. This paper emphasizes the examination of cooperative relationships within BVCs.

The second area of research focuses on digital trade rules. Meltzer (2016) provides a systematic overview of the development of US-style digital trade rules within the framework of RTAs. Zhou and Chen (2020) summarized the US-style digital trade rules under the framework of RTAs into seven representative US-style digital trade rules (duty-free electronic transmissions, free flow of cross-border data, nonmandatory localization of data storage, source code protection, liability exemption for Internet intermediaries, nonmandatory localization of technology, and nondiscriminatory treatment of digital products). They found that digital trade rules facilitate bilateral digital trade flows, with the trade-facilitating effects of free flow of cross-border data, noncompulsory localization of data storage, and source code protection rules being the most significant. Liu, Zhen, and Li (2021) found that regulatory convergence boosted digital trade growth by lower trade costs, improving bilateral network effects, and reducing institutional distance, with the film and music industry experiencing the greatest promotion effect. Li (2021) found that the depth of digital trade rules facilitated bilateral value-added trade, and the depth of service trade rules significantly and positively moderated the relationship between digital trade rules and value chain trade. These studies systematically classify and analyze digital trade rules, providing valuable methodological insights through categorical and quantitative approaches. However, they tend to focus primarily on changes in total value-added exports caused by trade rules, with no examination of the intensity of BVC collaboration. For instance, Li (2021) emphasizes the impact of digital trade rules on bilateral value-added trade volumes, whereas the BVC cooperation index developed in this study emphasizes the closeness of cooperative relationships between two countries.

2.2. Theoretical mechanism

This study also investigates the potential ways by which the depth of digital trade rules affects BVC cooperation. First, the depth of digital trade rules promotes BVC cooperation by lowering bilateral trade costs. Intermediate goods frequently cross international borders, making BVC cooperation a critical feature. Tariffs and trade inconveniences can potentially increase costs, significantly reducing the willingness to cooperate in the BVC. The "cost effect" of digital trade rules is evident in promoting trade facilitation, removing trade barriers between countries, and improving information-sharing efficiency in the trading process. Given that cross-border data flow is the foundation of digital trade, data localization measures in each country will increase the cost of using data and reduce the benefits of digital trade (Meltzer, 2019). Data flow provisions in digital trade rules lower the cost of information sharing (Goldfarb & Tucker, 2019) and reduce information asymmetry during the trade process (González & Jouanjean, 2017), improving supply and demand matching and facilitating BVC cooperation.

Following Viner's (1950) theory of static effects of regional economic integration, subsequent scholars have analyzed dynamic effects (e.g., returns to scale, returns to competition, and investment effects) (Meade, 1955). According to additional studies, trade rules positively impact bilateral foreign direct investment (FDI). For instance, Osnago, Rocha, and Ruta (2016) found that each additional clause in RTAs results in a 1.7 % increase in vertical FDI between countries. However, some scholars argue that the bilateral environment influences the effect of trade agreements on bilateral FDI. Carr, Markusen, and Maskus (2001) used the knowledge-capital model to combine horizontal and vertical FDIs and found that signing trade agreements reduced bilateral FDI flow.[5]

Bricongne, Franco Bedoya, and Lopez Forero (2022) suggested that FDI and trade have complementarity and substitutability to answer how FDI affects trade. On the one hand, multinational corporations with technological advantages may replicate their economics in foreign markets and locate production closer to foreign consumers via horizontal FDI to serve them more effectively. Multinational enterprises seek to offset trade costs (e.g., transportation, tariffs, or antidumping measures) through horizontal FDI (Carril-Caccia and Pavlova, 2020), which has a trade substitution effect. On the other hand, the FDI–trade correlation can also be positive. Multinational corporations use vertical FDI to increase production efficiency, which is more likely to occur when investing from developed to developing countries (Antràs & Yeaple, 2014). Vertical FDI allows firms to increase the scale of trade in intermediate goods, and therefore has a complementary effect on trade. Similarly, bilateral outward investment has a twofold effect on BVC cooperation. Horizontal FDI alters the original division of labor within the value chain, reducing trade in intermediate goods and significantly decreasing BVC trade. In contrast, vertical FDI establishes production and trade channels for intermediate goods between two countries, thereby increasing BVC trade. The relative magnitude of these two opposing effects determines the overall effect of bilateral FDI on BVC cooperation.

Third, the depth of digital trade rules strengthens BVC cooperation by increasing spatial knowledge spillovers. Numerous digital trade rules involve policies (e.g., intellectual property protection) that affect knowledge flows between countries (Jinji, Zhang, & Haruna, 2019; Liu & Zhen, 2022). Furthermore, because information spillover between exporters and importers significantly affects firms' export decisions (Fernandes & Tang, 2014), provisions on cross-border data flow in digital trade rules have the potential to reduce information-sharing costs and information asymmetry, provide firms with a valuable information platform, and facilitate knowledge spillover between firms (Chan & Hsieh, 2022), allowing firms' exports to increase. Furthermore, digital trade agreements facilitate international trade by lowering trade barriers, particularly for high-tech intermediate goods and ICT products. A country importing foreign intermediate goods with advanced technology can use "learning by doing" to promote its own technological upgrading (Antràs & Yeaple, 2014) and produce high-tech final goods for export, thereby enhancing BVC cooperation.

Fourth, the depth of digital trade rules promotes BVC cooperation by improving the bilateral business environment. High institutional quality demonstrates a pluralistic and inclusive political system that contributes to a level playing field, and reducing institutional distance between the two countries will boost bilateral trade (Álvarez, Barbero, Rodríguez-Pose, & Zofío, 2018). As the business environment becomes increasingly integrated into companies' daily operations, host governments frequently prioritize it as a key area of market environment reform (Lin & Zhang, 2019). Some provisions in the digital trade rules will hasten

[5] Carr et al. (2001) stated that when a developed country signs a trade agreement with another developed country, the decrease in horizontal FDI caused by the trade agreement's "cost-cutting effect" will outweigh the rise in vertical FDI due to the small skill gap. Ultimately, signing a trade agreement reduces bilateral FDI flows.

the improvement of the bilateral business environment, thereby closing the business environment gap. For example, in countries with higher digital barriers, the exemption for network intermediaries is comparatively low, which significantly increases the liability risk. Digital trade rules can send positive signals to the market by clarifying the exemption for network intermediaries. Accordingly, through the intermediary mechanism of differences in the business environment, digital trade rules can effectively reduce enterprises' market risks, lower the business's psychological defenses, and increase profitability, thus boosting the bilateral value-added trade.

In summary, the study proposes the following five hypotheses.

Hypothesis 1. *The depth of digital trade rules promotes BVC cooperation.*

Hypothesis 2. *Digital trade rules promote BVC cooperation through the "cost-cutting effect."*

Hypothesis 3. *The "investment effect" of digital trade rules improves BVC cooperation.*

Hypothesis 4. *The "spatial knowledge spillover effect" fosters BVC cooperation* via *digital trade rules.*

Hypothesis 5. *Digital trade rules enhance BVC cooperation by improving the ease of doing business.*

The relationship between RTAs and GVCs has been extensively studied; however, that between digital trade rules and GVCs has received relatively little attention. First, most existing research has focused on bilateral trade volume rather than value-added trade, and even less on the value-added trade of intermediate goods. Second, existing research primarily focuses on country-level effects, with few studies beginning at the industry level. This means that analyzing the trade promotion effect of digital trade rules frequently encounters endogenous issues such as reverse causality. Third, existing research has primarily focused on the effect of digital trade rules on export promotion, whereas a more interesting question is how digital trade rules affect BVC cooperation and the main paths of influence. Existing research fails to adequately address these aforementioned issues.

Compared with the existing literature, this study makes the following contributions. First, using 45 subsectors from the OECD TiVA database, it directly explores how the depth of digital trade rules affects BVC cooperation, thereby mitigating the endogeneity issue. Second, the effect of the digital trade rules depth on the degree of bilateral trade cooperation is examined in terms of value-added trade, and the regional heterogeneity of digital trade rules is considered. Third, the fundamental question of how the depth of digital trade rules affects party value chain cooperation is thoroughly studied.

3. Econometric specification and variable construction

3.1. Main regression

To analyze the correlation between the level of digital trade rules and BVC cooperation, this study uses the following econometric model:

$$lnBGVC_C_{ijkt} = \beta_0 + \beta_1 DTP_{ijt} + \beta_2 Controls + \delta_{it} + \delta_{jt} + \delta_{kt} + \delta_{ijkt} \quad (1)$$

where i, j, k and t denote importers, exporters, industries, and years, respectively.

$lnBGVC_C_{ijkt}$, the dependent variable, refers to the log of BVC cooperation level in industry k between countries i and, j in period t. DTP_{ijt} is the core explanatory variable, which represents the depth of digital trade rules signed by countries i and j in period t. It is measured by the depth of digital trade rules (*depth*) and the range of digital trade rules (*scope*), respectively, as in Section 3.2. *Controls* denotes control variables, including gravity model variables (e.g., GDP of countries i and j; geographic distance between countries i and j; and whether countries i and j share a common language, share a border, are colonies, and have a regional trade agreement). Furthermore, control variables control the relative position of value chains in countries i and j and the digital divide, innovation gap, and institutional quality gap between countries i and j. δs are a set of importer–time, exporter–time, and sector–time FEs. Country–time FE manages changes in the country over time, such as national demand shocks, supply shocks, and developing industries. Sector–time FE, which considers changes in industries over time, such as size and productivity, will impact bilateral industry value chain cooperation. This paper does not include country-pair FE because gravity model variables in the baseline model, such as common bilateral language, bilateral colonial relations, and whether bilateral territories are adjacent (all variables in the import–export dimension), would be covariates.

3.2. Variable construction

3.2.1. Depth of digital trade rules

Elsig and Klotz (2018) suggested that measuring the depth of digital trade provisions under the RTA framework involves the e-commerce chapter and other data-related chapters in RTAs (e.g., services, intellectual property and related ICT, big data, and data localization). Then, they summarized six indicators of the depth of digital trade provisions in RTAs, where the scope of digital trade rules emphasizes whether the provision is incorporated. The depth of the digital trade rule emphasizes the legal effect of the clause. Han, Cai, and Xu (2019) used the scope of digital trade rules in RTA to assess the importance of digital trade, where *Scope*1 denotes the number of digital trade regulations in RTAs and *Scope*2 denotes the number of words in the relevant digital trade regulations in RTAs. This study uses the proxy variables in accordance with Elsig and Klotz (2018) to measure the level of digital trade rules: the depth (*depth*) and scope of (*scope*) digital trade rules. Furthermore, the number of words in digital trade rules (*words*) was used as the main explanatory variable for robustness checks.

This study used Hofmann, Osnago, and Ruta's (2017) methodology to measure indicators of digital trade rules depth. The TAPED database at the University of Lucerne in Switzerland provided the original data for text quantification of RTA digital trade rules (Burri & Polanco, 2020). Moreover, the data are downscaled using the methodology proposed by Baier, Bergstrand, and Mariutto (2014), transforming RTAs involving more than two countries into RTAs between two countries, resulting in 710 country pairs. However, this method is better suited to measuring static indicators, which cannot accurately indicate the dynamic effects (Tie, Huang, & Xu, 2021). Ignoring the dynamic effects results in underestimating the two countries' economic and trade cooperation. Thus, this study refers to Tie et al. (2021) and utilizes the union set of RTA provision depth to reflect the depth between the two countries when multiple RTAs exist concurrently. It also treats the conclusion of RTAs following the first RTA as a reordering of RTAs between two countries. Fortunately, the TAPED database treats each RTA reformulation or expansion as a new RTA. If two countries signed multiple RTAs in the same year, this study selected the maximum depth to assess the depth of digital trade rules between the two countries in that year. If the content of digital trade terms in the same RTA is upgraded or changed, the maximum depth was used to determine the depth of the two countries' digital trade in that year.

The number of digital provisions included in an agreement from the TAPED database was used as a first measure of depth. In addition, an imperfect metric was the logical first step to capture the level of the depth of digital trade rules because the extent of policy commitments is dependent on the number of areas incorporated by an agreement (Laget et al., 2020). After normalization, the variable was defined as follows:

$$scope = \frac{\sum_{d=1}^{n} provision_{ijt}^{d}}{Provision} \# \quad (2)$$

where $\sum_{d=1}^{n} provision_{ijt}^{d}$ denotes the simple count of digital trade rules incorporated in the agreement between country i and j in year t. *Provision* presents the maximum number of total provisions that two countries can sign. For the sake of convenience in mathematical expression, we define $n = Provision$.

As an alternative measure of depth, we used the total depth of legally enforceable digital trade provisions incorporated in an agreement from the TAPED database, which assigns a value to each specific digital trade provision involved in an RTA in descending order of legal binding strength (Burri & Polanco, 2020). To be specific, for any RTAs, if the agreement does not contain the digital trade provision, the depth of digital trade rules ($depth_d$) takes the value of 0. If the agreement contains the digital trade provision, the depth of the digital trade rules ($depth_d$) is quantified based on legal validity or enforceability, with values assigned as 1, 2, or 3, respectively. In the case of identifying a clause in an agreement, if the provision is determined to be "soft" (i.e., emphasizing nonbinding obligations), $depth_d$ takes the value of 1. If the rule is "Hard," which means it contains the words "shall," "must," etc. (i.e., emphasizing binding obligations), the value of $depth_d$ is 3. If the rule is "mixed," which contains both binding and nonbinding obligations, the value of $depth_d$ is 2, as shown in Fig. 1. We defined the variable as follows:

$$depth = \frac{\sum_{d=1}^{n} depth_{ijt}^{d}}{Depth}\# \tag{3}$$

where $\sum_{d=1}^{n} depth_{ijt}^{d}$ is the sum of digital trade rules depth incorporated in the agreement between country i and j in year t, *Depth* is the maximum depth. Since the *Provision* presents the maximum number of total provisions that two country can sign, the maximum depth is three times of the value of *Provision*.

3.2.2. Bilateral value chain cooperation

Based on the research concept of Koopman, Powers, Wang, and Wei (2010), the dependence of trade in intermediate goods between two countries is used to measure the degree of BVC cooperation. In this study, we use Liu and Liu (2021) to construct the BVC cooperation index at the industry level in the following way:

$$BGVC_{Cijk} = \left(\frac{IV_{ijk}}{E_{ik}+E_{jk}} + \frac{FV_{ijk}}{E_{ik}+E_{jk}}\right) \times 100\# \tag{4}$$

where E_{ik} and E_{jk} represent the total export value of industry k in country i and country j, respectively. IV_{ijk} denotes the value-added exports of intermediate goods from country i to country j in sector k. $\frac{IV_{ijk}}{E_{ik}+E_{jk}}$ reflects the dependence of country j on intermediate goods in country i. FV_{ijk} represents the part of the value-added exports of industry k in country i which originates from country j, which presents the value-added exports of intermediate goods from country j to country i in sector k. $\frac{FV_{ijk}}{E_{ik}+E_{jk}}$ reflects the dependence of industry k in country i on intermediate goods of country j. The larger the index of BVC cooperation, the stronger the link between the two countries.

3.2.3. Control variables

GDP of country i and j ($lnGDP_i$, $lnGDP_j$) control the market size of country i and j. GDP is the gravity model's central variable, and value chain collaboration is typically more closely linked between economies with higher GDP. Therefore, this paper uses GDP from importing and exporting countries as control variables, with data sourced from CEPII's Gravity database. Geographic distance determines the "iceberg transportation cost," and the closer the geography, the lower the transportation cost (Krugman, 1991), referring to the new economic geography theory. This paper adds bilateral geographical distance ($lndist_{ij}$) as a control variable, using data from CEPII's Gravity database to promote value chain trade and cooperation between countries. The study introduces other gravity model variables (i.e., $Comlang_{ij}$, $Colony_{ij}$, $Contig_{ij}$). $Comlang_{ij}$ is a dummy variable that controls the presence of common languages between country i and j. In the benchmark model, the dummy variable for common languages equals 1 if the two countries speak the same language, and 0 otherwise. $Colony_{ij}$ is a dummy variable that takes the value of 1 if the two countries were in colonial relationship and 0 if not. The data are obtained from the Gravity database of CEPII. $Contig_{ij}$ is a binary variable that takes the value of 1 if the two countries are bordering, and 0 otherwise. The data are obtained from the Gravity database of CEPII.

In this study, we introduce the signed RTA dummy variable, which takes the value of 1 if both countries sign the RTA and 0 otherwise. Signing an RTA can reduce the uncertainty surrounding economic and trade policies, improving trade efficiency and fostering value chain cooperation. The value added from other member countries to total exports from RTA member countries increases, contributing to the expansion of value-added trade links between members (Orefice & Rocha, 2014). The information comes from the WTO database.

We include relative GVC position ($RGVC_P_{ijk}$) as a control variable in our model. There are two trends for industries to participate in GVC restructuring: one to improve their position in GVCs and another to increase GVC participation. The former will reduce profits in other countries (Yu, Zhou, Xie, & Wang, 2018). Therefore, the greater the difference between the two countries' relative positions in the GVC, the more likely trade frictions will arise. In this study, we follow Koopman et al. (2010) to measure a country's industrial position in the GVCs. In this study, we employ the absolute value of the difference between indirect value-added exports (IV) as a share of total exports (E) and the foreign value-added exports (FV) as a share of total exports (E) as an indicator of the relative GVCs position:

$$RGVC_{P_{ijk}} = \left| ln\left(1+\frac{IV_{ijk}}{E_{ik}+E_{jk}}\right) - ln\left(1+\frac{FV_{ijk}}{E_{ik}+E_{jk}}\right) \right| \times 100\# \tag{5}$$

The higher the value of $RGVC_P_{ijk}$, the more significant the gap in the relative positions of GVCs between country i and j in industry k, and the more likely the two countries will experience trade friction.

We also include three control variables: the bilateral Internet gap, bilateral innovation level gap, and bilateral institutional quality gap. First, the Internet is inextricably linked to digital trade. However, its global expansion is highly uneven, and the "digital divide" between the two countries is significant if the Internet penetration rates differ. When two countries have a significant "digital divide," they may increase bilateral trade to bridge the "digital divide" (Freund & Weinhold, 2004; Han et al., 2019). This study defines the bilateral "digital divide" as the difference in bilateral Internet penetration rates: $\text{digitalgap}_{ij} = \left|\text{lnternet}_{it} - \text{Internet}_{jt}\right|$. The data came from the WDI database. Second, this study calculates the bilateral innovation level gap: $rdgap_{ij} = \left|rd_{it} - rd_{jt}\right|$ using data from the WDI database, where rd_{it}, rd_{jt} denote the share of R&D expenditure in the GDP of country i and j, respectively. Third, the measure of institutional quality considers several dimensions, including the rule of law, perceptions of corruption, property rights, investor protection, and the political system, all of which can affect the economic development of a nation (Bankole, Osei-Bryson, & Brown, 2015). A narrowing institutional quality gap between two countries can increase bilateral trade (Álvarez et al., 2018). This study uses the Regulatory Quality subindex from the World Bank's Worldwide

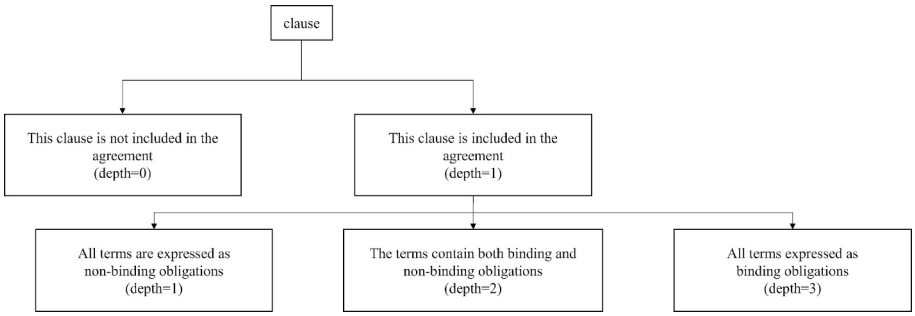


Fig. 1. Illustration of the assignment of values.

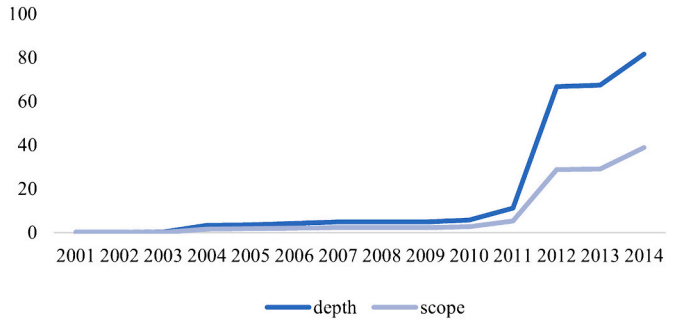


Fig. 2. Digital trade rules depth under regional trade agreement framework from 2001 to 2014.
(Source: TAPED database; author's own calculations.)

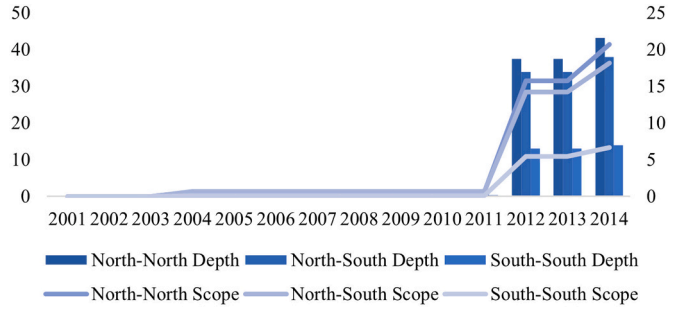


Fig. 3. Digital trade rules depth of different country pairs.
(Source: TAPED database; author's own calculations.)

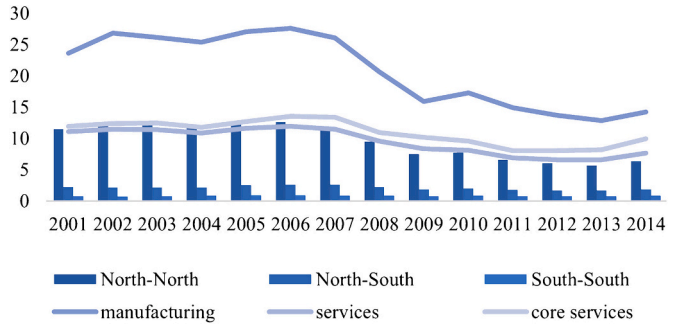


Fig. 4. Bilateral value chain cooperation.
Note: Variables are computed at the country-pair level and the industry level.
Source: UIBE Global Value Chain Database; author's own calculations.

Government Indicators (WGI) database[6] to measure the quality of nation-level institutions and defines bilateral institutional quality gap as follows: $IQgap_{ij} = |IQ_{it} - IQ_{jt}|$.

4. Data description

The US–Jordan Free Trade Agreement included digital trade rules for the first time in 2000, and they became effective in 2001. As shown in Fig. 2, the depth and scope of digital trade rules signed by countries between 2001 and 2010 are limited, and the rules are largely non-mandatory for these provisions. For example, the US–Korea Free Trade Agreement (KORUS), signed in 2007, is a prime example of US-style digital trade rules. Although the United States advocated for the "free flow of data across borders" in the agreement, it does not impose strong binding obligations on either party. Since 2011, the depth of digital trade rules signed by countries globally has increased significantly and continues to grow.

Fig. 3 depicts the signing of digital trade rules by various countries. Developed economies typically sign high-level digital trade rules, whereas less developed economies sign rules with a lower depth, with the depth of digital trade rules signed between developed and less developed economies being medium. Furthermore, the number of digital trade rules signed by "North–North," "North–South," and "South–South" countries show an increasing trend.

Fig. 4 illustrates the BVC cooperation level for various country pairs and sectors. The highest level of value chain cooperation is found in

Table 1
Descriptive statistics.

Variable	Obs.	Mean	Std. Dev.	Min	Max
lnbgvc_c	447,300	0.935	1.325	0	8.691
depth	447,300	0.063	0.118	0	0.405
scope	447,300	0.085	0.159	0	1.02
contig	447,300	0.079	0.27	0	1
comlang	447,300	0.065	0.246	0	1
colony	446,670	0.021	0.144	0	1
rta	447,300	0.736	0.441	0	1
lngdpi	447,120	19.087	1.65	14.38	23.075
lngdpj	446,040	18.893	1.69	14.38	23.58
lndist	447,300	7.801	1.1	4.088	9.856
digitalgap	447,255	10.268	5.884	0	57.733
rdgap	447,300	6.659	106.321	0	2013.75
IQ	447,300	5.595	98.482	0	2012.88
lnword	447,300	1.439	2.451	0	7.408
lnR	447,300	0.375	1.269	0	18.054
bit	447,300	0.169	0.374	0	1
wto	447,300	0.973	0.162	0	1

[6] Since 1996, the Worldwide Governance Indicators (WGI) project has reported aggregate and individual governance indicators for more than 200 countries and territories, encompassing six dimensions of governance (i.e., Voice and Accountability, Political Stability and Absence of Violence, Government Effectiveness, Regulatory Quality, Rule of Law, and Control of Corruption).

Table 2
Variable construction and data source.

Variable	Variable description and data source
$lnBGVC_C_{ijkt}$	Bilateral value chain cooperation level for sector k in countries i and j (*Source*: UIBE Global Value Chain database)
$depth_{ijt}$ $scope_{ijt}$	The depth ($depth_{ijt}$) and scope ($scope_{ijt}$) of digital trade rules signed by countries i and j (*Source*: TAPED database)
$contig_{ij}$	Whether countries i and j have territorial borders, dummy variable (*Source*: CEP II Gravity model database)
$comlang_{ij}$	Whether countries i and j share a common language, dummy variable (*Source*: CEP II Gravity model database)
$colony_{ij}$	Whether countries i and j have ever had a colonial relationship, dummy variable (*Source*: CEP II Gravity model database)
rta_{ijt}	Whether countries i and j have signed RTAs, dummy variable (*Source*: CEP II Gravity model database)
$lngdp_{it}$	The logarithmic of GDP of country i (*Source*: CEP II Gravity model database)
$lngdp_{jt}$	The logarithmic of GDP of country j (*Source*: CEP II Gravity model database)
$lndist_{ijt}$	The logarithmic of distance between country i and country j (*Source*: CEP II Gravity model database)
$digitalgap_{ijt}$	The digital divide between country i and country j (*Source*: World Bank database)
$rdgap_{ijt}$	The gap of innovation level between country i and country j (*Source*: World Bank database)
IQ_{ijt}	The gap of institutional quality between country i and country j (*Source*: World Bank database WGI indicators)

developed economies, followed by the level of value chain cooperation between developed and less developed economies, and finally between less developed economies. The degree of BVC cooperation between "North–North," "North–South," and "South–South" country pairs shows a significant downward trend in 2007–2008, followed by a brief upward trend in 2009–2010. The 2008 financial crisis may have triggered the bilateral economic fluctuations, resulting in a decrease in the level of BVC cooperation. Sector-wise, manufacturing has significantly more BVC cooperation than services. After subdividing the service sectors into core service sectors, the results show that the average degree of BVC cooperation among the core service sectors is greater than that of the overall service sectors.

Table 1 lists descriptive statistics for all variables, and Table 2 provides variable and data source definitions.

5. Empirical analysis and robustness check

5.1. Baseline results

This study conducts an econometric analysis using data from 2001 to 2014. Table 3 presents the baseline model's estimates. Columns (1)–(3) report the effect of the depth of digital trade rules on BVC cooperation, and columns (4) and (5) report the effect of the scope of digital trade rules on BVC cooperation. Columns (1) and (4) only contain the core explanatory variables, whereas columns (2) and (5) include gravity model variables such as importing and exporting countries' GDP, bilateral geographical distance, and common language dummy variable. Columns (3) and (6) include other control variables (e.g., the relative position of global value chain, bilateral digital divide, bilateral innovation level gap, and bilateral institutional quality gap). As shown in Table 3, the regression coefficients for the core explanatory variables are significantly positive in all regressions, which suggests that countries agreeing on digital trade rules will increase BVC cooperation. Thus, hypothesis 1 is confirmed.

The control variable estimation results show that the importing and exporting countries' GDP coefficients, binary colony variable and the digital divide, are all significantly positive. Meanwhile, the institutional quality gap is insignificant. The variables with significantly negative coefficients are bilateral geographical distance, common language binary variable, RTA binary variable, relative position in GVCs, and bilateral innovation level gap. According to the baseline results, market size, cultural similarity, and the bilateral digital divide facilitate BVC cooperation, whereas bilateral geographical distance, relative position of GVCs, RTA signing, and the bilateral innovation level distance impede BVC collaboration. The estimated results of *rta* are not as expected. This is due to the fact that the effect of RTA trade is dependent on the economic development, trade barriers, domestic business environment, and institutional differences of the contracting countries. Furthermore, the trade effect of the same trade agreement on different countries can be highly heterogeneous (Baier, Yotov, & Zylkin, 2019), and a negative effect can occur if the signed RTA does not reflect the level of economic development and competitiveness of enterprises in the contracting countries.

5.2. Robustness check

First, we run robustness tests by replacing the core variables. In this study, we test the effect of digital trade rules on the degree of BVC cooperation using two indicators of the depth and scope of digital trade rules in the benchmark regression, and now measure the core explanatory variable using the number of words of digital trade rules (*lnword*) in the RTAs text, and the results are shown in column (1) of Table 4. Subsequently, the results of this study are listed in columns (2) and (3) of Table 4, where the FE is changed and only the sector-year FE is included. Furthermore, this study uses Yang and Ai's (2021) methodology to assess the degree of BVC cooperation using foreign value added from direct importing countries. Among them, foreign value-added exports are divided into foreign value added directly from importing countries and third countries by source, which can be labeled as MVA_{ijkt} and OVA_{ijkt} for short, respectively. $TEXP_{ijkt}$ represents the total export value of industry k between country i and country j. Thus, Eq. (6) calculates the final BVC cooperation indicator:

$$R_{ijkt} = \frac{MVA_{ijkt}}{TEXP_{ijkt}}\# \tag{6}$$

We also consider that the international financial crisis has significantly affected GVC cooperation. To avoid perturbing the regression results caused by the mutation of the sample data during the international financial crisis, this study excluded sample data from 2008 to 2010 from the total sample data and then reregressed. Columns (6) and (7) of Table 4 present the regression results. As shown in the table, the estimated coefficients of the core explanatory variables remain significantly positive after excluding the sample data with possible perturbations.

The second topic is endogeneity. Almost all real-world economic and political factors that influence countries' decision to enter into digital trade agreements simultaneously impact BVC cooperation, making omitted variable bias a critical cause of endogeneity problems in empirical studies of RTAs using gravity models. Elsig and Klotz (2021) suggested that countries actively participating in the WTO's Electronic Commerce Cooperation Program discussions are more likely to include digital trade commitments in trade agreements. Furthermore, Baier et al. (2014) observed that the conclusion of trade agreements by contracting parties discriminates against noncontracting parties, causing them to conclude trade agreements, that is, a "domino effect." This study uses

Table 3
Digital trade rules depth and bilateral value chain cooperation.

lnBGVC_C	*depth*			*scope*		
	(1)	(2)	(3)	(4)	(5)	(6)
depth	**0.099*** **	**0.098*** **	**0.099*** **			
	(0.020)	**(0.020)**	**(0.020)**			
scope				**0.072*** **	**0.073*** **	**0.073*** **
				(0.015)	**(0.015)**	**(0.015)**
lngdpi		0.148***	0.149***		0.148***	0.149***
		(0.006)	(0.006)		(0.006)	(0.006)
lngdpj		0.041***	0.039***		0.041***	0.039***
		(0.001)	(0.001)		(0.001)	(0.001)
lndist		−0.074***	−0.073***		−0.074***	−0.073***
		(0.002)	(0.002)		(0.002)	(0.002)
comlang		−0.038***	−0.039***		−0.038***	−0.039***
		(0.008)	(0.008)		(0.008)	(0.008)
colony		0.075***	0.078***		0.075***	0.078***
		(0.012)	(0.012)		(0.012)	(0.012)
contig		0.439***	0.440***		0.439***	0.440***
		(0.007)	(0.007)		(0.007)	(0.007)
rta		−0.147***	−0.147***		−0.147***	−0.147***
		(0.005)	(0.005)		(0.005)	(0.005)
bgvc_p			−0.000***			−0.000***
			(0.000)			(0.000)
digitalgap			0.002***			0.002***
			(0.000)			(0.000)
rdgap			−0.000***			−0.000***
			(0.000)			(0.000)
IQ			0.000			0.000
			(0.000)			(0.000)
Constant	0.929***	−2.014***	−2.026***	0.929***	−2.015***	−2.026***
	(0.002)	(0.121)	(0.121)	(0.002)	(0.121)	(0.121)
Importer–time FE	Yes	Yes	Yes	Yes	Yes	Yes
Exporter–time FE	Yes	Yes	Yes	Yes	Yes	Yes
Sector–time FE	Yes	Yes	Yes	Yes	Yes	Yes
observations	447,300	445,230	445,185	447,300	445,230	445,185
R^2	0.566	0.577	0.577	0.566	0.577	0.577

Notes: Values in bold correspond to the coefficients and standard errors of the main explanatory variables of interest. Robust standard errors clustered at the industry level are in parentheses. *** $p < 0.001$, ** $p < 0.05$, * $p < 0.1$.

Osnago et al.'s (2016) approach[7] to measure the "domino effect." Accordingly, this study further introduces WTO member dummy variable (*WTO*), bilateral investment treaty dummy variable (*BIT*) (Laget et al., 2020) and "domino effect" (*third*, *diff*) based on Eqs. (1) and (2), and the estimation results are listed in columns (1)–(8) of Table 5. As shown in the table, the estimated coefficients of the core explanatory variables do not change substantially in comparison to the results of the baseline regression.

Reverse causality is another major cause of the endogeneity problem. We continue to consider how reverse causality affects the precision of the estimated coefficients of the depth of digital trade rules in four ways. In accordance with Tan and Qiu (2015), the BVC trade may affect the conclusion and implementation of bilateral digital trade rules, but it may not significantly affect the conclusion and implementation of multilateral or regional digital trade rules. All country pairs that agreed to bilateral digital trade rules during the sample period are excluded, and the benchmark model is reregressed. Columns (1) and (2) of Table 6 show the estimation results, which indicate that the estimated coefficients of the core explanatory variables remain significantly positive. Furthermore, the lagged one-period core explanatory variable is used as the core explanatory variable, as suggested by Nunn and Qian (2014), and the regression results are shown in columns (3) and (4) of Table 6. Moreover, the inverse of the logarithm of the distance between the two capitals is adopted as the instrumental variable. The reason for using instrumental variables is that when two countries are geographically close, cultural differences are smaller, bilateral political relations are more harmonious, and international trade agreements are easier to negotiate. As a result, the instrumental variables significantly correlate with the depth of digital trade rules. Furthermore, the distance between the two capitals is a historical variable that does not significantly affect current BVC cooperation and meets the instrumental variable's exogeneity condition. The inclusion of an Importer–Exporter FE in the gravity model can effectively address the endogeneity issue (Baier & Bergstrand, 2007). Therefore, this FE is included in this paper, and the regression results are shown in columns (7) and (8), which are consistent with the baseline regression results.

5.3. Heterogeneity analysis

The first topic is country heterogeneity. Herein, pairs between developed countries are labeled "North–North" country pairs, matched pairs between less developed countries are labeled "South–South" country pairs, and matched pairs between developed and less developed countries are labeled "South–North" pairs.[8] Table 7 presents the

[7] This study measures the "domino effect" by adding the number of free trade agreement (FTAs) signed between country *i* and non-*j* countries plus the sum of the number of FTAs signed between country *j* and non-*i* countries (*third*). In addition, the study uses the absolute difference between the number of FTAs signed by country *i* and non-*j* countries and the number of FTAs signed by country *j* and non-*i* countries to reflect the difference in the number of FTAs signed by the two economies (*diff*).

[8] Because OECD countries have more developed economies, this study classifies them as North, and non-OECD countries as South. OECD countries: Australia, Austria, Belgium, Canada, Switzerland, Czech Republic, Germany, Denmark, Spain, Finland, France, UK, Greece, Hungary, Ireland, Italy, Japan, Korea, Lithuania, Luxembourg, Latvia, Mexico, Netherlands, Norway, Poland, Slovakia, Slovenia, Sweden, USA; non-OECD countries: Croatia, Cyprus, Malta, Romania, Brazil, China, Bulgaria, Indonesia, and India.

Table 4
Robustness check.

lnBGVC_C	*lnword*	FE change		R_{ijkt}		Financial crisis	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
lnword	**0.007*** (0.001)**						
depth		**0.160*** (0.023)**		**0.097** (0.031)**		**0.128*** (0.022)**	
scope			**0.102*** (0.017)**		**0.073** (0.023)**		**0.094*** (0.016)**
Importer–time FE	Yes	No	No	Yes	Yes	Yes	Yes
Exporter–time FE	Yes	No	No	Yes	Yes	Yes	Yes
Sector–time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Constant	1.937*** (0.121)	2.008*** (0.213)	2.012*** (0.213)	0.590*** (0.147)	0.590*** (0.147)	2.088*** (0.135)	2.088*** (0.135)
Observations	445,185	140,870	140,870	445,185	445,185	349,470	349,470

Notes: The values in bold represent the coefficients and standard errors of the main explanatory variables of interest. Robust standard errors clustered at the industry level are in parentheses. *** $p < 0.001$, ** $p < 0.05$, * $p < 0.1$. Control variables are consistent with those in the baseline regression.

Table 5
Omitted variable bias.

lnBGVC_C	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
depth	**0.099*** (0.020)**	**0.099*** (0.020)**	**0.099*** (0.020)**	**0.099*** (0.020)**				
scope					**0.073*** (0.015)**	**0.073*** (0.015)**	**0.074*** (0.015)**	**0.074*** (0.015)**
WTO	0.086*** (0.018)	0.087*** (0.018)	0.087*** (0.018)	0.086*** (0.018)	0.085*** (0.018)	0.087*** (0.018)	0.087*** (0.018)	0.086*** (0.018)
BIT		−0.071*** (0.004)	−0.074*** (0.004)	−0.075*** (0.004)		−0.071*** (0.004)	−0.074*** (0.004)	−0.075*** (0.004)
third			−0.027*** (0.002)	−0.026*** (0.002)			−0.027*** (0.002)	−0.026*** (0.002)
diff				0.004*** (0.001)				0.004*** (0.001)
FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
constant	−2.051*** (0.121)	−2.053*** (0.121)	−1.971*** (0.121)	−1.981*** (0.121)	−2.051*** (0.121)	−2.053*** (0.121)	−1.971*** (0.121)	−1.981*** (0.121)
observations	445,185	445,185	445,185	445,185	445,185	445,185	445,185	445,185

Notes: The values in bold correspond to the coefficients and standard errors of the main explanatory variables of interest. Robust standard errors clustered at the industry level are in parentheses. *** $p < 0.001$, ** $p < 0.05$, * $p < 0.1$. Control variables are consistent with those in the baseline regression. Importing country–time FE, exporting country–time FE, and sector–time FE are added.

Table 6
Endogeneity problem discussion.

lnBGVC_C	Exclude bilateral data		Lagged one period		Instrumental variable		Importer–Exporter FE	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
depth	**0.105*** (0.023)**				**0.098*** (0.029)**		**0.102*** (0.018)**	
scope		**0.082*** (0.017)**				**0.077*** (0.022)**		**0.074*** (0.013)**
L.depth			**0.095*** (0.022)**					
L.scope				**0.071*** (0.016)**				
Kleibergen–Paap rk LM Statistics					2.5e+04 [0.0000]	2.2e+04 [0.0000]		
Kleibergen–Paap rk Wald F Statistics					4.6e+04 {16.38}	6.1e+04 {16.38}		
Importer–time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Exporter–time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sector–time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Importer–Exporter FE	No	No	No	No	No	No	Yes	Yes
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	347,715	347,715	413,280	413,280	413,280	413,280	445,185	445,185
R^2	0.589	0.589	0.579	0.579	0.027	0.027	0.676	0.676

Notes: The values in bold represent the coefficients and standard errors of the main explanatory variables of interest. Robust standard errors clustered at the industry level are in parentheses; *** $p < 0.001$, ** $p < 0.05$, * $p < 0.1$. Control variables match those in the baseline regression. Importing country–time FE, exporting country–time FE and sector–time FE are added. Endogenous variable *depth* for columns (1)–(3) and *scope* for columns (4)–(6). *P*-values are in brackets and critical values at the 10 % level of the Stock-Yogo test are in curly brackets.

Table 7
Country heterogeneity.

lnBGVC_C	North–North		South–South		North–South	
	(1)	(2)	(3)	(4)	(5)	(6)
depth	**0.180*** **		**−0.205*** **		**0.107*** **	
	(0.031)		**(0.048)**		**(0.028)**	
scope		**0.135*** **		**−0.161*** **		**0.072*** **
		(0.023)		**(0.036)**		**(0.021)**
FE	Yes	Yes	Yes	Yes	Yes	Yes
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes
Constant	3.434***	3.434***	−4.203***	−4.203***	−2.750***	−2.750***
	(0.219)	(0.219)	(0.447)	(0.447)	(0.159)	(0.159)
Observations	203,895	203,895	63,270	63,270	178,020	178,020

Note: The values in bold represent the coefficients and standard errors of the main explanatory variables of interest.
Robust standard errors clustered at the industry level are in parentheses. *** $p < 0.001$, ** $p < 0.05$, * $p < 0.1$. Control variables match those in the baseline regression. Importing country–time FE, exporting country–time FE, and sector–time FE are added.

regression results. The regression results show that the trade effects of the depth of digital trade rules differ significantly in the "North–North," "South–South," and "North–South" country pairs. In particular, in the "North–North" country pairs (columns (1) and (2) in Table 7) and "North–South" country pairs (columns (5) and (6) in Table 7), the depth of digital trade rules improves BVC cooperation. The rules of digital trade among developed countries have a greater impact on promoting BVC cooperation compared to those among less developed countries because developed countries' economic models and industrial structures are very similar and there are few cultural differences. Furthermore, the demands on digital trade rules are more consistent. Most of the developed countries in the sample are European countries, with similar claims on sensitive provisions (e.g., those related to cross-border data flow and personal privacy), making it easier to sign in-depth digital trade rules involving frontier issues. As a result, digital trade's full potential is utilized. In "South–South" country pairs, the depth of digital trade rules makes BVC cooperation difficult. This study contends that less developed countries frequently experience issues with their digital technology infrastructure, and their domestic legal systems for protecting intellectual property are imperfect, which can impede BVC cooperation when high-standard digital trade rules are signed.

Second, regional heterogeneity is examined, focusing on the effect of digital trade rules signed by member countries of two major economic cooperation organizations, Asia–Pacific Economic Cooperation (APEC) and the EU,[9] on BVC cooperation. Columns (1) and (2) of Table 8 show the regression results for APEC, while columns (3) and (4) show the regression results for EU. The digital trade rules agreed upon by the APEC member countries impede BVC cooperation, whereas the digital trade rules agreed upon by the EU member countries facilitate BVCs. The digital trade rules include more "behind-the-border" provisions and domestic regulatory issues (Ahmed, 2019); hence, the significant differences in economic development levels and cultural differences among the APEC member countries will limit the effect of the digital trade rules on trade promotion. By contrast, the EU member states share similar levels of economic development, language, and culture. As a result, signing digital trade rules will considerably facilitate BVC cooperation.

Third, we analyze the differences in the effect of various types of digital trade rules on BVC cooperation. This study categorizes digital trade provisions into three categories (Table 9) based on Peng, Yang, and Shen's (2021) approach and the TAPED database)[10]: trade promotion, data-related, and privacy protection and cybersecurity provisions.

Trade promotion provision is primarily associated with comprehensive environmental clauses that encourage the development of digital trade facilitation. Implementing clauses that address specific issues raised in the WTO's e-commerce work plan, such as developing domestic legal and regulatory frameworks for e-commerce, paperless trade, and electronic authentication, can effectively reduce regulatory barriers between countries in digitalized trade and services. This strategy also improves the overall level of digital trade facilitation.

Data-related provision aims to include topics that have not yet been discussed within the framework of the WTO Working Group on Electronic Commerce and focuses on more controversial and cutting-edge topic areas such as cross-border data flows, data localization prohibitions, and nondisclosure of software source code. Data flow has taken on critical significance in determining whether digital trade can be successfully conducted (González & Jouanjean, 2017). Trade and production in the digital economy are dependent on it. To be more specific, the United States supports the free flow of data. From the CPTPP to the USMCA to the US–Japan Digital Trade Agreement, the US has emphasized the importance of "digital" and data, while targeting policies and regulatory measures that impede cross-border data flows. The EU has advocated for a balance between data freedom and personal privacy. China and other emerging markets are focusing on cybersecurity.

Privacy and cybersecurity provisions are associated with the domestic regulatory environment and impact digital service exports. Low levels of data privacy protection can undermine trust between digital businesses and consumers, limiting digital service exports (Guellec & Paunov, 2018). Conversely, stringent data privacy protection can increase operational costs for digital businesses, put them at a competitive disadvantage internationally, and reduce the competitiveness of digital service exports. The privacy protection and cybersecurity provisions include intellectual property protection, consumer data protection, and cybersecurity.

The estimation results in Table 8 reveal that the impact of the digital trade rules' subdivision provisions on the degree of BVC cooperation is heterogeneous. The trade promotion and data-related provisions have a significant positive effect on BVC cooperation, whereas the privacy

[9] APEC members include Australia, Brunei Darussalam, Canada, Chile, the People's Republic of China, Hong Kong, China, Indonesia, Japan, the Republic of Korea, Malaysia, Mexico, New Zealand, Papua New Guinea, Peru, the Philippines, the Russian Federation, Singapore, Thailand, the United States of America, and Vietnam. The data for Chinese Taipei is missing, so it is not included. EU members include Austria, Belgium, Bulgaria, Croatia, Cyprus, Czechia, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, the Netherlands, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, Sweden, and the United Kingdom. The UK left the EU on January 31, 2020, and the article's sample period is 2001–2014, so the UK is included.

[10] Peng et al. (2021) construct the "trade promotion provisions," "data-related provisions," and "privacy protection provisions" using the TAPED database, which the University of Lucerne provided in January 2021. "Privacy protection provisions" include personal data protection as well as the responsibility of the online intermediary platform. This study constructs "privacy protection and network security provisions" by adding IP protection and network security-related provisions to "privacy protection provisions."

Table 8
Regional and provision heterogeneity.

lnBGVC_C	APEC		EU		Provision heterogeneity	
	(1)	(2)	(3)	(4)	(5)	(6)
Depth	**−0.274*** **		**0.214*** **			
	(0.072)		**(0.027)**			
Scope		**−0.226*** **		**0.166*** **		
		(0.053)		**(0.020)**		
TPdepth					**0.156*** **	
					(0.044)	
DPdepth					**0.097** **	
					(0.033)	
PCPdepth					**−0.379*** **	
					(0.060)	
TPscope						**0.225*** **
						(0.034)
DPscope						**0.023**
						(0.015)
PCPscope						**−0.451*** **
						(0.059)
FE	Yes	Yes	Yes	Yes	Yes	Yes
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes
Constant	1.006***	1.008***	−0.004	−0.004	−2.016***	−1.948***
	(0.338)	(0.338)	(0.199)	(0.199)	(0.121)	(0.121)
Observations	47,160	47,160	254,655	254,655	445,185	445,185

Note: The values in bold represent the coefficients and standard errors of the main explanatory variables of interest.
Robust standard errors clustered at the industry level are in parentheses. *** $p < 0.001$, ** $p < 0.05$, * $p < 0.1$. Control variables match those in the baseline regression. Importing country–time FE, exporting country–time FE, and sector–time FE are added.

protection and cyber security provisions have a significant negative impact. The main reasons are that the trade promotion provision is based on the WTO framework and promotes bilateral value-added trade by lowering trade barriers, simplifying trade processes, and specifying tariff exemptions for electronic commerce. Furthermore, the trade promotion provision addresses fewer sensitive issues, making it easier for contracting parties to achieve deeper cooperation. Thus, trade promotion provisions have a stronger positive impact on BVC trade. Data flows are an important factor in bilateral cooperation regarding data provisions. Measures such as data localization, however, increase the cost of accessing and using data and reduce the advantages of digital trade (Meltzer, 2019). Both parties can reduce the uncertainty that firms face in trade, improve data utilization, and thus increase BVC cooperation by signing digital trade provisions that clarify data flow regulations. Data-related provisions are frequently considered highly economically, politically, and ideologically sensitive because they involve "frontier issues," which are currently highly controversial. It is difficult for countries to achieve deep cooperation on data flows, reducing the role of digital trade provisions in promoting BVC cooperation. A stricter level of protection for privacy protection and cybersecurity provision, which involve data protection, can limit data flow and increase the cost of bilateral trade. Lower levels of privacy protection may pose cybersecurity risks. Herein, privacy protection and cybersecurity clauses impede BVC cooperation.

Fourth, we examine industry heterogeneity. The subsample regressions in this study are based on the OECD's classification of manufacturing, services, and core services.[11] The estimation results of Table 10 show that digital trade rules have a significant positive promotion effect on BVC cooperation in core services but no effect on BVC cooperation in manufacturing and services. The reason is that the core service industries primarily include finance, insurance, and other business services. The cross-border data flows are closely linked to bilateral financial and other business activities. The bilateral agreement on digital trade rules can effectively promote cross-border data flow, which has a greater impact on BVC cooperation in core services.

6. Mechanism analysis

6.1. Cost-cutting effect

The costs of bilateral trade have always significantly affected the global division of production. Lower bilateral trade costs result in more intermediate commodity trade between economies, directly encouraging BVC cooperation. Bilateral trade costs are calculated using the logarithm of bilateral trade cost data from the ESCAP-World Bank Trade Cost database. Herein, we introduce the interaction term of bilateral trade cost and the depth of digital trade rules; the regression results are shown in columns (1) and (2) of Table 11. The estimated coefficients of bilateral trade costs are significantly negative, which suggests that they impede BVC cooperation. The interaction term's coefficient is significantly positive, which suggests that obtaining comprehensive digital trade regulations between the two countries can reduce costs and thus

[11] Core services include publishing activities; motion picture, video and television program production, sound recording and music publishing activities; programming and broadcasting activities; telecommunications; computer programming, consultancy and related activities; information service activities; financial service activities, except insurance and pension funding; insurance, reinsurance and pension funding, excluding compulsory social security; activities auxiliary to financial services and insurance activities; real estate activities, legal and accounting activities, head office activities, and management consulting activities; Architectural and engineering activities; technical testing and analysis; scientific research and development; other professional, scientific, and technological activities; veterinary activities; and administrative and support services.

Table 9
Digital trade provisions types.

First-level indicators	Second-level indicators	Third-level indicators	Specific indicators
Trade promotion provision	WTO rules complementary deepening	Applicability of WTO rules	1.10. Does the agreement mention the applicability of WTO rules to e-commerce?
		E-commerce Tariffs	1.11.1. Is there a provision on the nonimposition of custom duties? 1.11.2. Does the agreement include a provision on custom value of carrier mediums?
		Nondiscriminatory treatment of E-commerce	1.3. Does the agreement provide for national treatment in e-commerce? 1.4. Does the agreement provide for MFN treatment in e-commerce? 1.7. Does the e-commerce chapter refer to provisions on other chapters of the agreement, such as and investment chapters? 1.8. Are there services (investment) market access and national treatment commitments for the sectors needed for e-commerce?
	Trade facilitation provisions	Establishing an e-commerce framework	1.16. Does the agreement include provisions that recognizes the importance of promoting or facilitating electronic commerce? 1.17. Does the agreement include provisions for facilitating the use of e-commerce by SMEs?
		Paperless Trade	1.19. Does the agreements include a provision on paperless trading?
		Electronic Authentication	1.20. Does the agreement include provisions on electronic authentication, electronic signatures, or digital certificates?
Data-related provision	Cross-border data flows	Cross-border data flows	1.28.1. Do data flows in e-commerce chapters include the provisions on data flows? 1.28.2. Do data flows in e-commerce chapters include a mechanism to address barriers to data flows? 1.28.3. Do data flows in e-commerce chapters include a provision imposing data localization requirements? 1.28.4. Do data flows in e-commerce chapters include provisions on banning or limiting data localization requirements? 2.1. Does the agreement include a provision on the free movement of the data? 3.1. Is there any reference to the transfer of data flows in the telecommunications chapter/provisions? 3.2. Computer and related services (CRS) 3.3. Audiovisual 3.4. Financial service
	Data localization	Data localization	2.3. A provision imposing data localization requirements? 2.4. Provisions on banning or limiting data localization requirements?
	Disclosure of software source code	Disclosure of software source code	1.33. Does the agreement include prohibitions to require the transfer of or access to the source code of software owned by a person as a condition of the import, distribution, sale, or use of such software?
	IP	IP protection	1.22. *E*-commerce IP 4.4 Does the agreement include provisions on the duration of the terms of protection of copyright and related rights? 4.12. Does the agreement include provisions on Internet Domain names?
Privacy protection and cybersecurity provision	Data protection	Data protection	1.23. Provisions on consumer protection 1.24. Data protection
	ISP liability	ISP liability	4.13. Does the agreement include provisions on the liability of Internet Service providers? 4.14. Does the agreement include provisions on the safe harbors for Internet Service providers?
	Cybersecurity	Cybersecurity	1.32. Does the agreement include provisions on cybersecurity?

Source: The TAPED Database; Author integration and classification.

promote BVC cooperation.[12] Thus, hypothesis 2 is confirmed.

6.2. Bilateral investment effect

An important factor that affects GVCs is FDI. To measure bilateral FDI in this study, we used data from the UNCTAD database. According to Burbidge, Magee, and Robb (1988), we use the Inverse Hyperbolic Sine Transformation to process FDI investment flows to avoid many missing zeros and negative values.[13] Columns (3) and (4) of Table 11 show that the estimated coefficient of bilateral FDI is significantly negative. One possible explanation is the substitutability of horizontal FDI and trade. Multinational corporations with technological advantages enter foreign markets via horizontal FDI to reduce trade costs and reliance on trade. The estimated coefficient of the interaction term between bilateral FDI and the depth of digital trade rules is significantly positive, indicating that digital trade rules lower bilateral trade barriers and reduce contracting parties' willingness to engage in horizontal FDI, while the bilateral skill difference narrows with continuous knowledge spillover, further reducing contracting parties' horizontal FDI (Jang, 2011). Therefore, the depth of digital trade rules encourages BVC cooperation via the "bilateral investment effect," which supports our hypothesis 3.

[12] The mechanism of the effect of the depth of digital trade rules on bilateral value chains can also be comprehended by deriving the following equation:

$$\frac{\partial lnbgvc_c}{\partial lncost} = \alpha_1 + \lambda_1 depth(\alpha_1 < 0, \lambda_1 > 0\ )\#(7)$$

$\alpha_1 < 0$ suggests that bilateral trade costs inhibit BVC cooperation. The greater the depth of digital trade rules, the lower the bilateral trade costs and the less of an inhibitory effect they have on BVC cooperation.

[13] The specific transformation method is as follows:

$$FDI = ln\left(fdi + \sqrt{fdi^2 + 1}\right)\# \quad (8)$$

Table 10
Industry heterogeneity.

lnBGVC_C	Manufacturing		Services		Core Services	
	(1)	(2)	(3)	(4)	(5)	(6)
depth	**0.034**		**0.029**		**0.148****	
	(0.041)		**(0.027)**		**(0.045)**	
scope		**0.054**		**0.010**		**0.138*****
		(0.030)		**(0.020)**		**(0.035)**
FE	Yes	Yes	Yes	Yes	Yes	Yes
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes
Constant	−2.320***	−2.322***	−1.950***	−1.949***	−2.178***	−2.181***
	(0.217)	(0.217)	(0.150)	(0.150)	(0.228)	(0.228)
Observations	158,288	158,288	187,967	187,967	69,251	69,251

Notes: The values in bold represent the coefficients and standard errors of the main explanatory variables of interest. Robust standard errors clustered at the industry level are in parentheses. *** $p < 0.001$, ** $p < 0.05$, * $p < 0.1$. Control variables are consistent with those in the baseline regression. Importing country–time FE, exporting country–time FE, and sector–time FE are added.

Table 11
Mechanism Analysis.

lnBGVC_C	*tradecost*		*FDI*		*spillover*		*DB*	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
depth	−0.243***		0.065		0.059**		−0.048	
	(0.045)		(0.084)		(0.022)		(0.038)	
scope		−0.190***		0.082		0.038*		−0.033
		(0.034)		(0.059)		(0.017)		(0.030)
tradecost	−0.099***	−0.099***						
	(0.002)	(0.002)						
FDI			−0.055***	−0.055***				
			(0.016)	(0.016)				
spillover					0.017***	0.016***		
					(0.000)	(0.000)		
DB							0.001***	0.001***
							(0.000)	(0.000)
depth × *tradecost*	**0.082*****							
	(0.009)							
scope × *tradecost*		**0.063*****						
		(0.007)						
depth × *FDI*			**0.178*****					
			(0.053)					
scope × *FDI*				**0.130*****				
				(0.039)				
depth × *spillover*					**0.012*****			
					(0.003)			
scope × *spillover*						**0.011*****		
						(0.002)		
depth × *DB*							**0.001*****	
							(0.000)	
scope × *DB*								**0.001*****
								(0.000)
FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Constant	−1.476***	−1.474***	−4.428***	−4.428***	−2.093***	−2.093***	−2.099***	−2.100***
	(0.121)	(0.121)	(0.215)	(0.214)	(0.121)	(0.121)	(0.139)	(0.139)
Observations	443,970	443,970	59,085	59,085	445,185	445,185	349,470	349,470

Notes: The values in bold represent the coefficients and standard errors of the main explanatory variables of interest. Robust standard errors clustered at the industry level are in parentheses. *** $p < 0.001$, ** $p < 0.05$, * $p < 0.1$. Control variables are consistent with those in the baseline regression. Importing country–time FE, exporting country–time FE, and sector–time FE are added.

6.3. Spillover effect of spatial knowledge

Knowledge spillover results from research collaboration, personnel contact, and imitation demonstration effects among firms. This study uses the number of R&D personnel as a proportion of the total population in both countries to measure spatial knowledge spillover for mechanism analysis, drawing on Liu and Zhen's (2022) research idea. The data are obtained from the World Bank database, and the estimation results are shown in columns (5) and (6) of Table 11. The estimated spatial knowledge spillover coefficients are significantly positive, implying that spatial knowledge spillover promotes BVC cooperation. The coefficient of the interaction term is significantly positive, suggesting that the depth of digital trade rules reduces information asymmetry and information-sharing costs of multination enterprises and facilitates positive network externalities and knowledge spillovers (Chan & Hsieh, 2022), thereby promoting BVC trade. As a result, the depth of digital trade rules can promote BVC cooperation via the "spatial knowledge spillover effect," confirming hypothesis 4.

6.4. Effect of the business environment

The World Bank's Doing Business report includes several important aspects of the regulatory environment that affect domestic firms in the respective economy.[14] The Doing Business database provides the Ease of Doing Business Score for each economy to evaluate its business environment on a scale of 0–100, with higher scores indicating a relatively convenient business environment. To determine the bilateral business environment variables (*DB*), *DB* is defined as follows: $DB_{ij} = DB_i + DB_j$; the greater the indicator, the easier the business will be conducted in country *i* and *j*, and vice versa. The result is listed in columns (7) and (8) of Table 11. The estimated coefficient of *DB* is significantly positive, suggesting that greater ease of doing business in both countries will facilitate value chain cooperation. The interaction term's coefficient is significantly positive, which suggests that the depth of digital trade rules can encourage BVC cooperation by improving bilateral ease-of-influence conditions. Accordingly, hypothesis 5 is confirmed.

7. Conclusion

Digital trade rules drive the development of a new generation of trade rules. Moreover, negotiations in various countries on digital trade rules are gradually shifting to "frontier issues," such as the free flow of cross-border data, data localization, and data protection. Whether signing high-level digital trade rules promote BVC cooperation is a question with significant policy implications.

To examine the relationship between the depth of digital trade rules and the degree of BVC cooperation, we use the UIBE GVC and TAPED databases. We also perform rich heterogeneity analysis (i.e., country pairs, regional, industry, and clause). We obtain the following findings. For example, signing deep digital trade rules will increase BVC cooperation. After accounting for endogeneity issues and various measurement indicators, the promotion effect of the depth of digital trade rules on BVC cooperation remains robust. Second, the depth of digital trade rules primarily promotes BVC cooperation via four channels (i.e., cost-cutting effect, investment effect, spillover effect of spatial knowledge, and business environment improvement effect). Third, the depth of digital trade rules between developed economies and between developed and less developed economies facilitates BVC cooperation. Digital trade rules between less developed economies harm BVC cooperation. Fourth, the depth of digital trade rules encourages BVC collaboration in core services. However, it does not significantly impact BVC cooperation in manufacturing value chain or bilateral services value chain. Fifth, digital trade rules that promote trade and data-related digital trade rules positively affect BVC collaboration. In contrast, privacy protection and cyber security digital trade rules harm BVC cooperation.

In conclusion, our findings highlight the importance of deep digital trade rules in enhancing BVC cooperation. To encourage such cooperation, policymakers should carefully consider the characteristics of their negotiating partners. Furthermore, it is crucial to consider the potential impact of each provision on core service industries while maintaining a balanced approach to privacy protection and cybersecurity provisions within digital trade rules.

This study has certain limitations. First, due to the limitations of the database, the data used in this study is not up-to-date. Second, owing to data sample constraints, we are unable to delve into the specific impact mechanism of digital trade rules on BVC cooperation between less developed economies. Understanding the impact mechanism of digital trade rules on BVC between less developed economies would facilitate the promotion of such cooperation and enhance our comprehension of the impact mechanism. Additionally, the rapid development of artificial intelligence and blockchain technologies has challenged the existing digital trade rules, potentially rendering the digital trade rules examined in this study obsolete or insufficiently comprehensive. Given the rapidly evolving digital trade landscape, it is crucial to investigate the effects of emerging digital trade rules on BVC cooperation.

[14] The World Bank's Doing Business report includes quantitative indices for starting a business, obtaining construction permits, obtaining electricity, registering property, obtaining credit, protecting minority investors, paying taxes, trading internationally, enforcing contracts, and dealing with bankruptcy.

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Data availability

No

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Glossary

Bilateral value chain (BVCs): *Bilateral value chains refer to international production sharing between two countries, where production is divided into activities and tasks that are carried out exclusively in these two countries. In this paper, we use an index to measure the degree of BVC cooperation.:*
Digital trade: *According to the IMF, OECD, UNCTAD, and the WTO, in the second edition of the Handbook on Measuring Digital Trade (2023), digital trade was defined as "All international trade that is digitally ordered and/or digitally delivered".:*
Foreign direct investment (FDI): *According to the United Nations Conference on Trade and Development, foreign direct investment (FDI) is defined as an investment involving a long-term relationship and reflecting a lasting interest and control by a resident entity in one economy (foreign direct investor or parent enterprise) in an enterprise resident in an economy other than that of the foreign direct investor (FDI enterprise or affiliate enterprise or foreign affiliate).:*
Foreign value added: *According to the United Nations Conference on Trade and Development, Foreign value added (foreign value added as a share of exports) indicates what part of a country's gross exports consists of inputs that have been produced in other countries.:*
Global value chains (GVCs): *According to the United Nations Industrial Development Organization, Global value chains (GVCs) refer to international production sharing, a phenomenon where production is broken into activities and tasks carried out in different countries.:*
Regional trade agreements (RTA):*According to Word Bank Group, a regional trade agreement (RTA) is a treaty between two or more governments that define the rules of trade for all signatories.:*